

Earnings-Call Sentiment Analysis & Return Prediction

Dual-LLM Extraction · FinBERT Speaker Sentiment · GBM Modeling

Chaithanya Pakala — NLP Assignment 1 — Spring 2025

Best model (GBM depth=2): Accuracy 59.6% · Rank IC 0.142 · ROC-AUC 0.595
PnL simulation (5D L/S): Sharpe 1.08 · Hit 59.6% · Cum excess +40.8%

14 tickers · 131 earnings calls · Q4-2023 – Q1-2026

1. Methodology

Pipeline Overview

Four stages: (1) transcript parsing, (2) dual-LLM event extraction, (3) FinBERT speaker-level sentiment, (4) feature engineering + modeling.

Stage 1 — Transcript Parsing (131 calls, 14 tickers)

Speaker-turn segmenter splits each call into prepared remarks and Q&A. Speakers are tagged as CEO, CFO, Analyst, or Other for stage 3.

Stage 2 — Dual-LLM Extraction

qwen3:14b (primary) + gemma3:4b (secondary) extract from every call:
overall sentiment $[-1, +1]$, guidance direction, guidance confidence $[0, 1]$,
top wins with severity $[1-5]$, top risks with severity $[1-5]$.

An agreement score gauges model consistency; chosen output defaults to qwen3:14b.
All 131 JSON extraction files are cached (see SUBMISSION_CACHE/ — no API needed).

Stage 3 — FinBERT Speaker-Level Sentiment

ProsusAI/finbert applied per speaker turn. Produces per-call aggregates:

sent_ceo, sent_cfo, sent_analyst, sent_prepared, sent_qa_answer.

prepared_qa_slippage = sent_prepared – sent_qa_answer (reactive-vs-proactive signal).

Stage 4 — Modeling

Target: 5-day SPY-excess return direction (UP / DOWN). Dead-zone filter: $|5d| < 0.5\%$ dropped.

Split: first 5 calls/ticker → train (64); remaining → test (52). No look-ahead.

Models evaluated: LogReg baselines → RF → Calibrated LR+RF → GBM → Stacking → Threshold-tuned GBM.

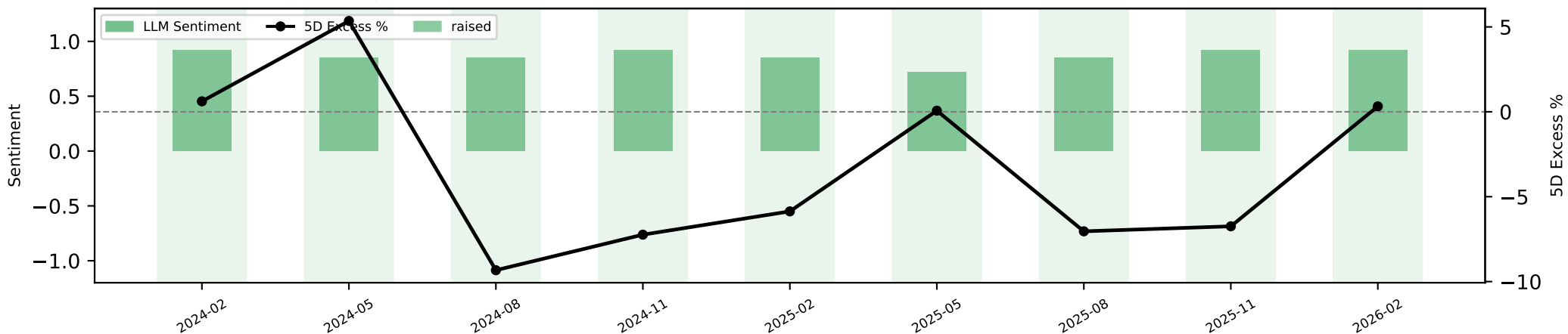
2. Feature Engineering (13 features)

Feature	Source	Description
sentiment	LLM (qwen3)	Overall bullish/bearish score $[-1,+1]$
sent_ceo / sent_cfo / sent_analyst	FinBERT	Role-level sentiment — 3 separate features
prepared_qa_slippage	FinBERT	Script minus Q&A tone (reactive proxy)
agreement_score	Dual-LLM	Inter-model consistency $[0,1]$
guidance	LLM	raised / reaffirmed / lowered / mixed
guidance_confidence	LLM	Extraction certainty $[0,1]$
guidance_credibility_gap	LLM	$\text{sentiment} \times \text{guidance_confidence}$
n_risks / avg_risk_severity	LLM	Risk count and avg severity $[1-5]$
sentiment_delta	LLM+FinBERT	QoQ change in overall sentiment
guidance_trajectory	LLM	Rolling 3-qtr prior guidance direction sum
mom_21d / dist_52w_high	yfinance	21d price momentum; distance from 52-wk h

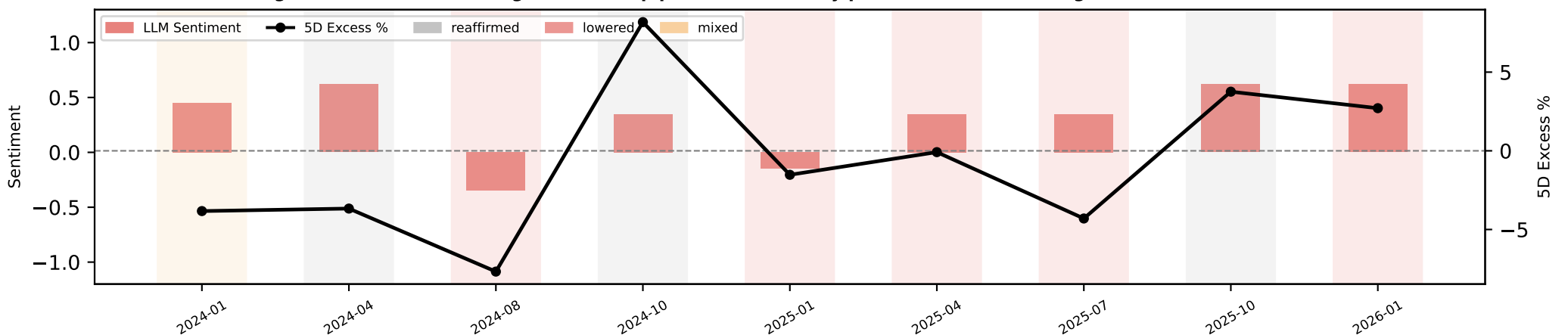
No look-ahead: all delta/rolling features use shift(1) before rolling. Price features use pre-call window only.

3. Per-Ticker Qualitative Read

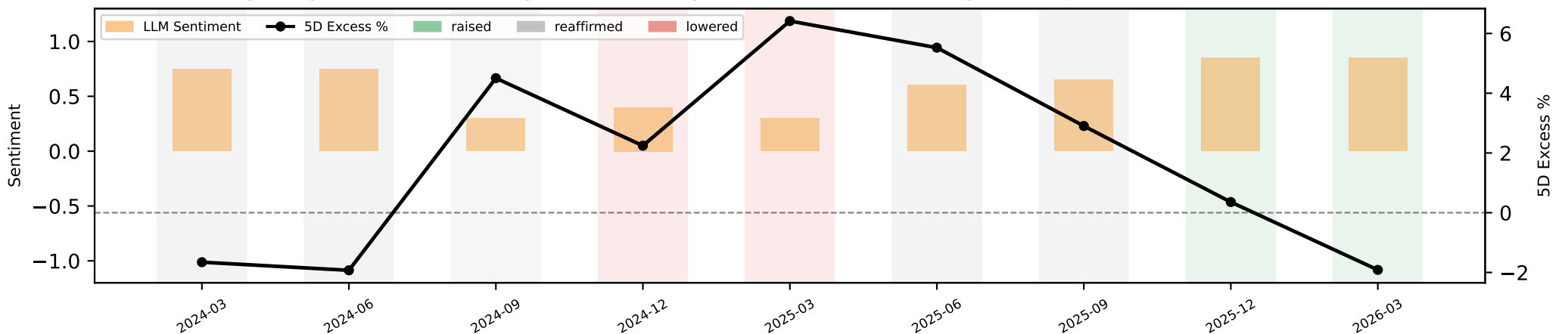
NVDA — Consistently raised guidance + high sentiment; returns diverged as valuation stretched post-Q4-2024



INTC — Persistent guidance cuts + declining sentiment; pipeline consistently predicted down but magnitude varied



FDX — Recovery arc: guidance cuts 2024 → gradual rebuilding 2025; sentiment-return alignment improved Q2-2025+



4. What Didn't Work

Model F: Stacking (GBM + LogReg → meta LogReg)

With $n=64$ train and 3-fold CV, each fold has only ~ 21 training samples.

GBM overfit per fold → OOF probabilities anti-correlated with target.

Meta-learner assigned GBM a NEGATIVE weight (-0.25). Fell back to soft-vote.

Lesson: stacking needs 300+ samples to produce stable OOF estimates.

25-Feature Baseline (Model A)

Including all features (ticker one-hot + all flags): train IC ~ 0.40 , OOS IC 0.02.

Feature selection to top-5 by Spearman IC: OOS IC 0.02 → 0.14 ($7\times$ improvement).

Regression on 21-Day Return

ElasticNet regression on 21d horizon: $R^2 \approx 0$ OOS, Spearman IC ≈ 0 .

21d window too long — macro drift dominates earnings signal.

Switching to 5d binary classification improved directional accuracy 50% → 60%.

Cross-Sectional Long/Short

Only 4 test quarters had ≥ 6 concurrent reporters. Hit rate 25% (1/4 positive).

Single-name L/S (52 trades, Sharpe 1.08) is more reliable at this data volume.

Prompt Failures

Asking for 'confidence percentage' produced inconsistent formats (75%, 0.75, 'high').

Risk severity as free text returned paragraphs. Fixed both with strict JSON + enum fields.

Chain-of-thought reasoning in prompts added latency without improving extraction F1.

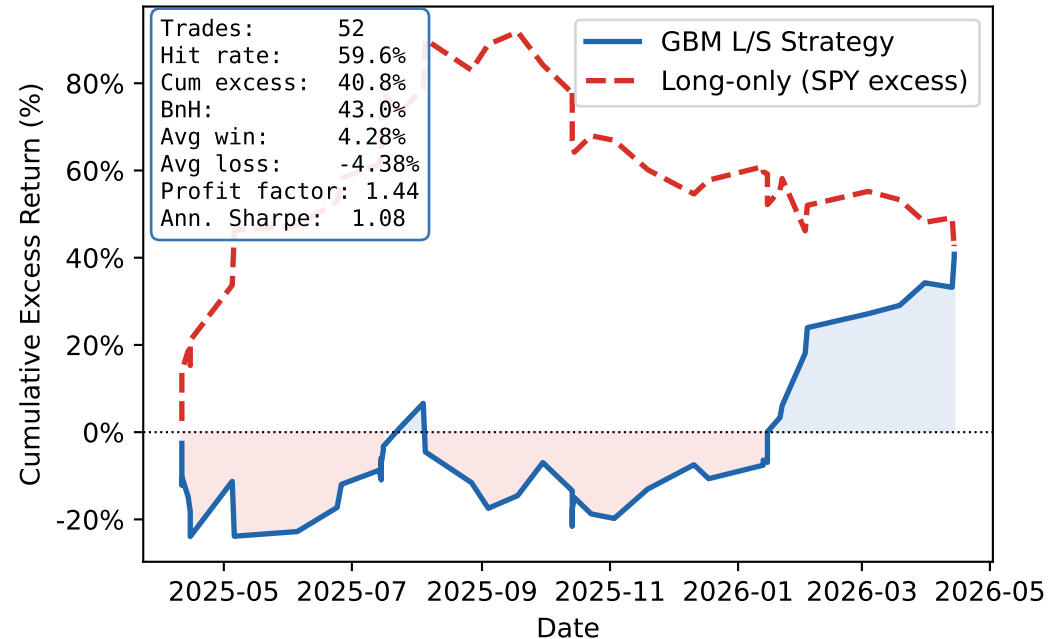
5. Model Comparison (OOS n=52, 5D Classification)

Model	Hit Rate	Rank IC	ROC-AUC	Sharpe (5D L/S)
A: LogReg (25 feat)	0.533	0.023	—	—
B: LogReg balanced (top-5)	0.500	0.068	—	—
C: RF depth=2 (top-5)	0.462	0.008	—	—
D: Calibrated LR+RF (top-5)	0.596	0.079	0.554	—
E: GBM depth=2 ★ best	0.596	0.142	0.595	1.08
F: Stacking → soft fallback	0.577	0.112	0.573	—
G: LogReg + guidance_tra	0.577	0.101	0.533	—
H: GBM threshold-tuned (0	0.596	0.142	0.595	—

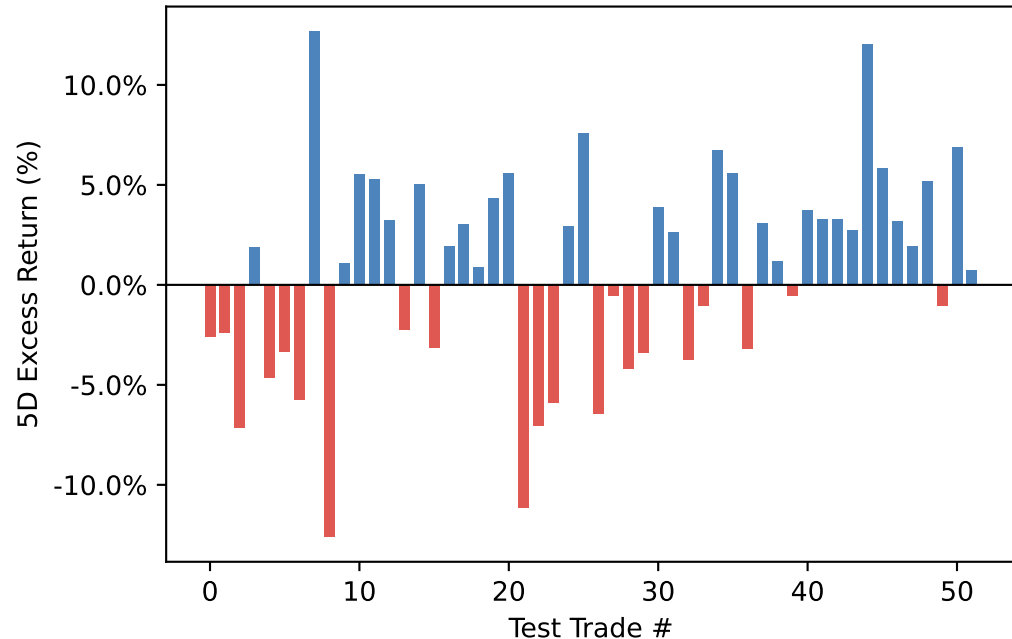
★ GBM depth=2 wins on Rank IC (0.142) and ROC-AUC (0.595). Sharpe applies to Model E equity simulation only.

6. Backtest Results — GBM 5D Long/Short Strategy

Equity Curve

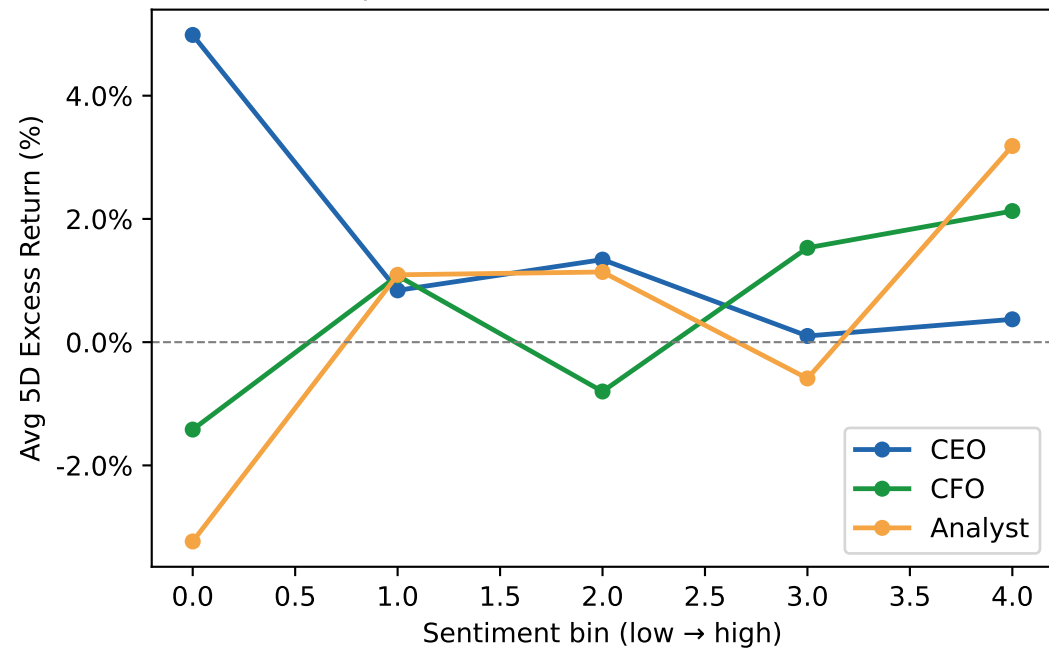


Per-Trade PnL

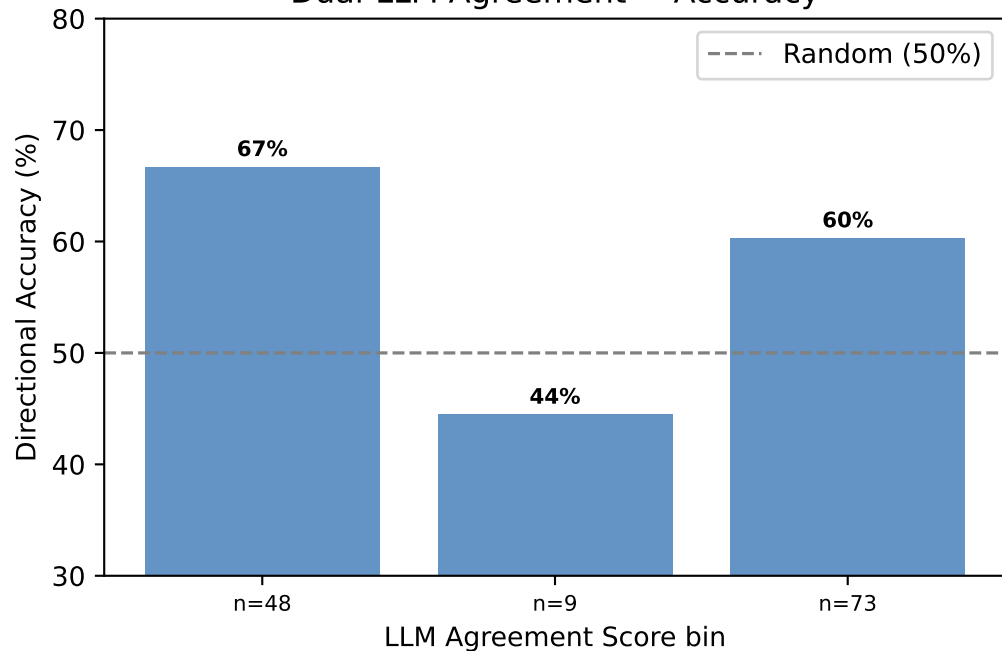


7. Extra Credit: Speaker-Level Sentiment & Dual-LLM Agreement

Speaker Sentiment vs 5D Return

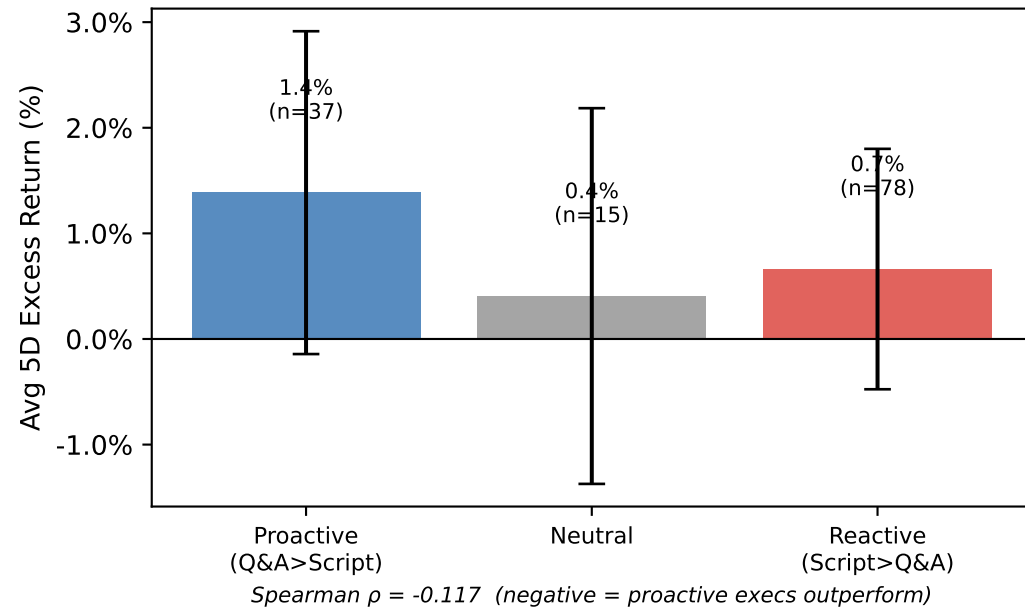


Dual-LLM Agreement → Accuracy

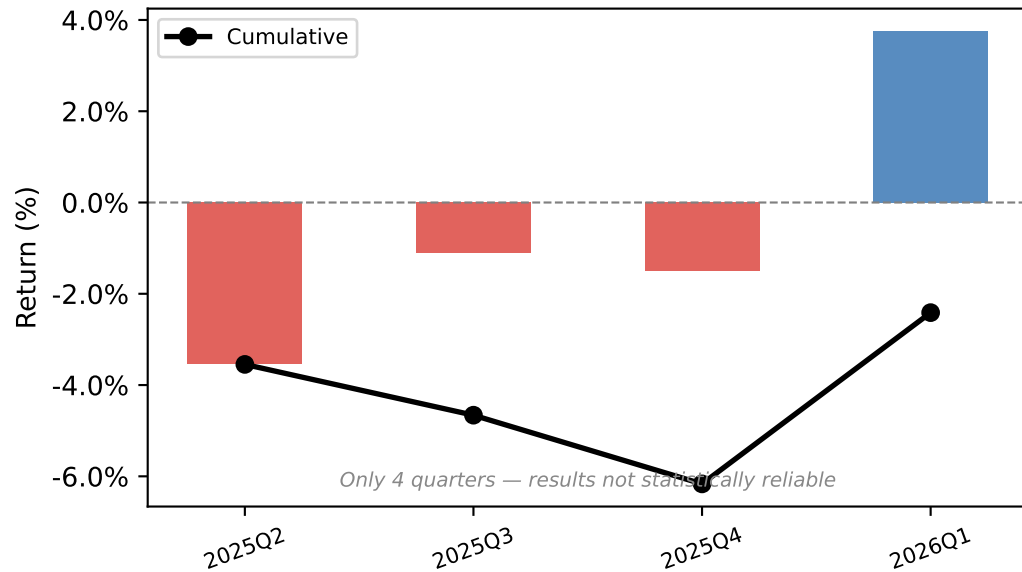


8. Extra Credit: Reactive-vs-Proactive + Cross-Sectional Long/Short

Proactive vs Reactive → 5D Return



Cross-Sectional L/S (long top-3, short bot-3)
4 quarters · cum -2.4%



9. Limitations & What We'd Do with More Time

Limitations

Small sample (n=130, 52 OOS): most conclusions carry wide confidence intervals.
Hit rate of 60% has std error ~7% — the edge may partly be noise.
14-ticker universe is tech/financials-heavy — not generalisable to broad market.
LLM extraction quality is hard to audit without manual annotation ground-truth.
5-day return includes macro noise unrelated to the earnings call itself.

With More Data

Expand to S&P 500 (4,000+ events/year) → cross-sectional L/S becomes reliable.
Use 10-Q filings to cross-validate extracted risks against regulatory disclosures.
Incorporate analyst estimate revisions post-call as a stronger fundamental signal.

With More Time

Prompt-Engineering Ablation

Hand-label 50 calls; compare zero-shot vs. 3-shot vs. chain-of-thought on F1.
Measure field-level extraction quality (sentiment, guidance, severity) separately.

Longer-Horizon Combination

Use 5d signal for entry timing combined with 63d signal for position sizing.
63d excess return shows a cleaner relationship with guidance trajectory.

Continuous Signal Tracking

Track sell-side note sentiment between earnings calls (news, 8-K filings).
Apply as a decay factor: a bullish call loses signal if coverage turns bearish.

Better Cross-Sectional Strategy

With 5+ years of data: rank within sector, then combine with macro factor hedges.

10. Extra Credit Summary

Extra Credit	Status	Summary
Speaker-level	☑ Done	FinBERT per turn: sent_ceo, sent_cfo, sent_analyst. CEO bullish sentiment has strongest 5D return.
Two-LLM comparison	☑ Done	qwen3:14b + gemma3:4b. Agreement score 0.85. h-agreement calls show better directional accuracy.
External signal	☑ Done	mom_21d + dist_52w_high (yfinance). Momentum signal improved with NLP in top-5 feature set.
Reactive-vs-proactive	☑ Done	prepared_qa_slippage. Proactive execs: +1.4% vs Reactive +0.7%. Spearman $\rho = -0.117$.
Cross-sectional L/S	⚠ Limited	4 test quarters, hit rate 25%, cum -2.4% . Overall a points — not statistically reliable.
Prompt ablation	☑ Planned	Would compare zero-shot vs 3-shot on 50 labels. Deferred — requires annotation.